AI Agent protocol for multi-modality

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https://www.ietf.org/archive/id/draft-hw-protocol-agent-00.html

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Background

- The Agent protocols have garnered significant attention across diverse technical domains. Agent protocols are being introduced by open-source and standardization organizations, e.g., A2A protocol by Google [1], Agent protocol proposals in IETF [2] [3] and ETSI [4].
- **As an example, for mobile telecommunications,** 3GPP has also proposed to incorporate Agent capabilities within the future 6G network.
  - Numerous use cases for Agentic scenarios have been adopeted within the 3GPP SA1 working group [5].
  - The 3GPP SA2 working group has already started to study how the 6G network can support Agentic network and Agent communication [6].
  - Additionally, the 3GPP CT working groups are organizing studies on AI Agent protocols for the 6G network. In this context, IETF AI Agent protocols are highly recommended as a reference [7].

With the advancement of AI technologies, AI Agent traffic (e.g., Token communication, multiple AI Agent communication & cooperation) will account for the majority of network traffic.

End-to-End Agent protocols are needed to better support AI Agent traffic transmission and ensure the quality of such transmission, e.g., based on the characteristics of AI Agent traffic.

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Multi-modality of an AI Agent Traffic

- An AI Agent traffic may consist of multiple “Part” with multi-modal data (e.g., text, file (raw data, URL), sensor data, structured data, video, audio and image) [3].
- Different “Part” of an AI Agent traffic may be delivered with different transmission modes (e.g., Streaming, Push Notification, Burst).

4.1.7. Artifact

Artifacts represent task outputs.

Field	Type	Required	Description
`artifactId`	`string`	Yes	Unique identifier (e.g. UUID) for the artifact. It must be unique within a task.
`name`	`string`	No	A human readable name for the artifact.
`description`	`string`	No	Optional. A human readable description of the artifact.
`parts`	array of `Part`	Yes	The content of the artifact. Must contain at least one part.
`metadata`	`object`	No	Optional. Metadata included with the artifact.
`extensions`	array of `string`	No	The URIs of extensions that are present or contributed to this Artifact.

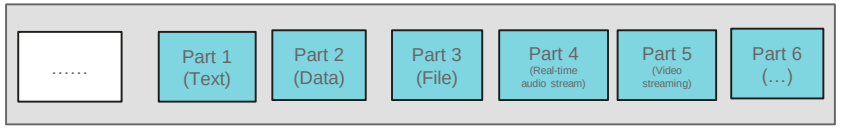
4.1.6. Part

`Part` represents a container for a section of communication content. Parts can be purely textual, some sort of file (image, video, etc) or a structured data blob (i.e. JSON).

Field	Type	Required	Description
`text`	`string`	Optional (OneOf)	The string content of the `text` part.
`raw`	`bytes`	Optional (OneOf)	The `raw` byte content of a file. In JSON serialization, this is encoded as a base64 string.
`url`	`string`	Optional (OneOf)	A `url` pointing to the file's content.
`data`	`any`	Optional (OneOf)	Arbitrary structured `data` as a JSON value (object, array, string, number, boolean, or null).
`metadata`	`object`	No	Optional. metadata associated with this part.
`filename`	`string`	No	An optional `filename` for the file (e.g., "document.pdf").
`mediaType`	`string`	No	The `media_type` (MIME type) of the part content (e.g., "text/plain", "application/json", "image/png"). This field is available for all part types.

Note: A `Part` MUST contain exactly one of the following: `text`, `raw`, `url`, `data`

@[A2A] example: an AI Agent traffic with multi-modal “Parts”.



[A2A] https://a2a-protocol.org/latest/specification/#416-part
[3] https://www.ietf.org/archive/id/draft-hw-ai-agent-6g-00.html

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Gap: current AI Agent protocols

Header

```
POST /message:send HTTP/1.1
Host: agent.example.com
Content-Type: application/a2a+json
Authorization: Bearer token
```

Payload

```
{
 "message":{
  "role": "ROLE_USER",
  "parts": [
   {
    "text": "Analyze this image and highlight any faces."
   },
   {
    "raw": "iVBORw0KGgoAAAANSUhEUgAAAAUA..."
    "filename": "input_image.png",
    "mediaType": "image/png",
   }
  ],
  "messageId": "6dbc13b5-bd57-4c2b-b503-24e381b6c8d6"
 }
}
```

Text part

Raw file part

Scenario: Client sends an image for analysis, and the agent returns a modified image.
→ Request with Text & Image with raw File Upload.

- E.g., different multi-modal "Parts" are encapsulated in the **HTTP protocol payload.**
- The underlaying network nodes cannot distinguish the different "Parts" and cannot know the traffic characteristics, unless they deeply decapsulate the payload.

Limitation: Content of different "Parts" are encapsulated within lower layer protocol payload (e.g., as payload of HTTP protocol, MoQ protocol, QUIC protocol), the underlying network (e.g., mobile network, transport network) remains unaware of the distinct Parts and their individual requirements, such as specific QoS needs.

@[A2A]: https://a2a-protocol.org/latest/specification/#67-file-exchange-upload-and-download

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Gap: current underlaying network

Interaction between Agent Client and Agent Server via underlying Network:

```
+--------+                  +--------+
| Agent  |                  | Agent  |
| Client |                  | Server |
+--------+                  +--------+
    |                            |
    |       +------------+       |
    +-------| Underlying |-------+
            |  Network   |
            +------------+
```

- Mobile network (e.g., 3GPP 6G network),
- Wired network (e.g., Transport network, WiFi)
- Etc.

Indistinguishable Multi-modality of AI Agent Traffic.

■ **Two ways to transmit AI Agent traffic**

1. The multi-modal data of an AI Agent traffic is encapsulated into multiple lower layer streams/connections (e.g., HTTP streams/Connections, MoQ streams, QUIC Streams/connections):
   - the underlying network lacks awareness that these streams belong to the same AI Agent traffic. Consequently, while network nodes (e.g., core network user plane function, radio access network base stations) can guarantee the QoS for individual stream during data forwarding, the overall performance (e.g., latency) for the traffic may be reduced and the network resource cost may increase.
2. The multi-modal data of an AI Agent traffic is encapsulated within a single lower layer stream/connection, this stream/connection exhibits inherent multi-modal characteristics:
   - This manifests as significant variance in Part sizes (e.g., text snippets vs. video files) and divergent quality of service (QoS) criticality across Parts, e.g., regarding transmission urgency and latency tolerance thresholds.

Limitation: the underlying network (e.g., mobile network, transport network, wired network, WiFi) remains unaware of the distinct Parts and their individual requirements, such as specific QoS needs.

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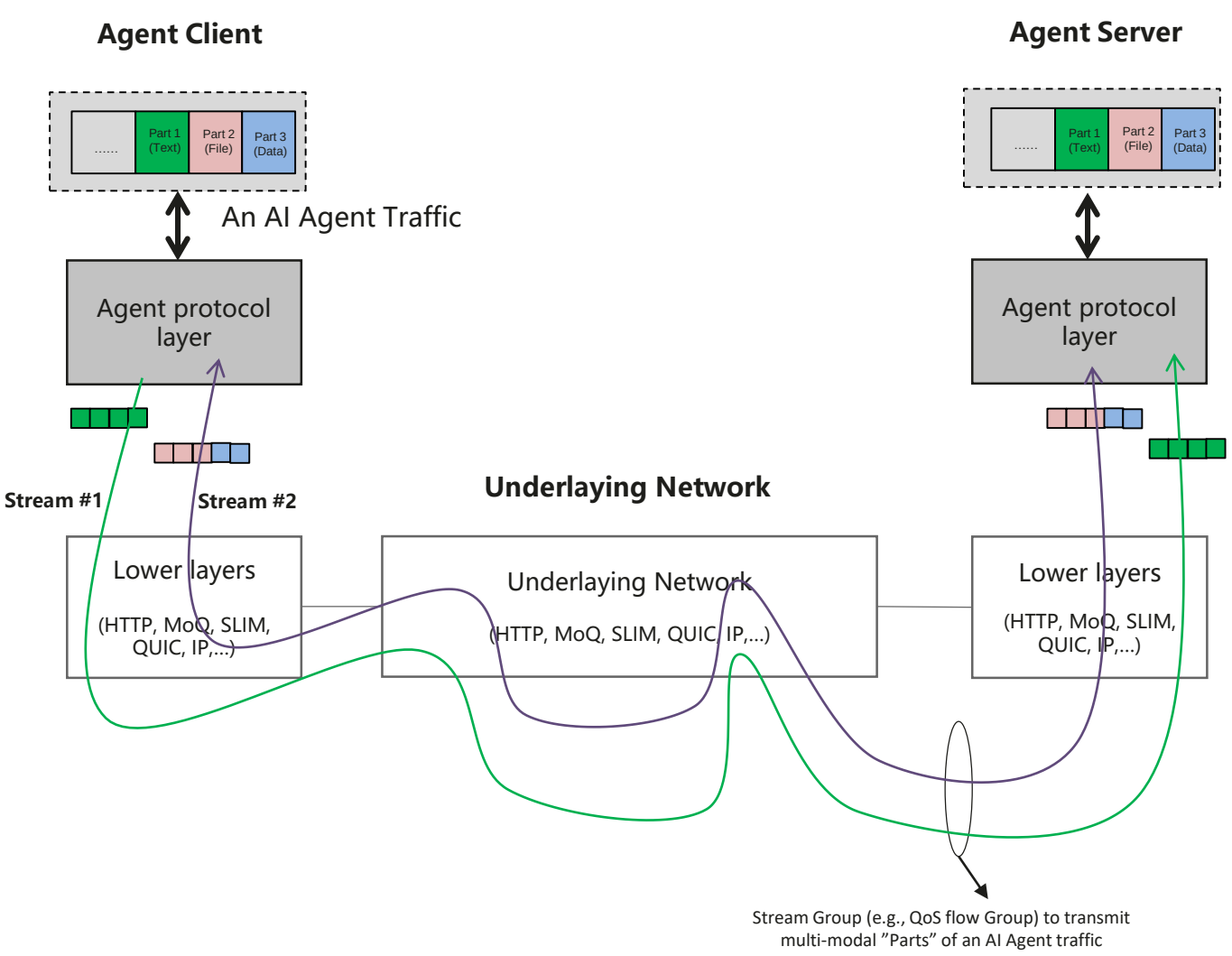
Underlaying network for multi-modal Agent traffic



- The underlaying network can aware of the distinct traffic characteristics and QoS requirements of the different "Parts" of an AI Agent traffic.
- Different transmission treatment policies are configured and bound to the different multi-modal data of an AI Agent traffic based on their different traffic characteristics and QoS requirement (e.g., delay budget, priority, error rate, reliability, data rate, packet loss).
- The requirements of both the individual "parts" and the whole AI Agent traffic are satisfied , to increase the performance and the network resource usage.
  - Taking mobile network as an example, the AI Agent client and server need to notify necessary information on its AI Agent traffic to the underlying network for better performance.
  - Based on this, the mobile network provisions and configures the corresponding QoS enforcement policies.
  - Subsequently, the mobile network maps incoming packets of application traffic to the appropriate QoS Flows by matching against the enforcement policies and the traffic information (e.g., IP 5-tuple, metadata).

AI Agent protocol for multi-modal Agent traffic

```
+-----------------------------------------------------+
|                    Agent Protocol                   |
|               {Correlation Information}             |
+-----------------------------------------------------+   --+
| e.g., HTTP Protocol       | e.g., MoQ               |     |
| {Correlation Information} | {Correlation Information}|     |
+-----------------------------------------------------+     |-- Lower Layers
|                   Transport Protocol                |     |
|               {Correlation Information}             |     |
+-----------------------------------------------------+   --+
```

Figure 2. Enhancements in Agent protocol layer and/or lower layers.

```
+-----------------------------------------------------+
| POST /rpc/ HTTP/2.0                                 |
+-----------------------------------------------------+
| Host: agent.example.com                             |
| Content-Type: application/json                      |
| Content-Length: xxx                                 |
| Correlation Information: Relationship indictor      |
+-----------------------------------------------------+
```

Figure 3. ***Correlation Information*** *in the HTTP header.*

An inter-stream relationship indicator, as an example of the correlation information, is introduced to explicitly signal that multiple distinct streams belong to the same AI Agent traffic.

4.1. Multiple Streams/connections for Multi-modal Agent Traffic

- A source Agent (client or server) maps & transmits multi-modal data of an AI Agent traffic into distinct lower layers streams/connections based on modality and QoS requirements.
  - For example, a source Agent transmits multi-modal data of an AI Agent traffic into distinct HTTP streams. Each HTTP stream is characterized by a unique IP 5-tuple and carries data of only one modality.
  - As another example, multi-modal data of an AI Agent traffic is transmitted via distinct transport layer streams/connections (e.g., MoQ streams, QUIC streams/connections, TCP connections)
- The source Agent needs to notify the mapping information to the underlying network to make alignment. Based on the mapping information, the underlying network can identify the correlation of the different lower layer streams, and execute dedicated QoS guarantee accordingly.
- Moreover, it is possible that the Agent protocol layer and/or lower layers need some enhancements, e.g., to encapsulate AI Agent assistance information (e.g., traffic characteristics, multi-modal data types, correlation information).

The AI Agent protocols in the Agent client and server should have the capabilities and features to perform the actions above**, e.g., mapping multi-modal data of Agent traffic to different lower layer streams, notifying the mapping relationship to the underlying network (e.g., via control plane of mobile network), and notifying necessary information to lower layers** to enable the latter to encapsulate Agent traffic assistance information.

The capabilities and features of the AI Agent protocol should be highlighted and standardized, e.g., in IETF working group(s).

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AI Agent protocol for multi-modal Agent traffic

```
+---------------------------+-------------+-------------+-------------+
|Header{                    |             |             |             |
|   Number of Parts,        |             |             |             |
|   {Part 1,                |             |             |             |
|      {Traffic information, |             |             |             |
|       QoS requirements},  |   Payload   |   Payload   |     ...     |
|    Part 2,                |   (Part1)   |   (Part2)   |             |
|      {...},               |             |             |             |
|   ...                     |             |             |             |
|   }                       |             |             |             |
|}                          |             |             |             |
+---------------------------+-------------+-------------+-------------+
```

*Figure 4. Supplementary metadata in the **header**.*

For example, the header contains fields such as:

- the total number of Parts,
- sequence number of each Part and the corresponding traffic information (e.g. size of a Part, modality of a Part, and transmission mode of a Part), QoS requirements (e.g., latency, error rate, priority, privacy), etc.

The contents of different Parts will be encapsulated in the payload in sequence with the same order of the supplementary metadata in the header.

4.2. Signal Streams/connections for Multi-modal Agent Traffic

- the source Agent can encapsulate multi-modal data of an AI Agent traffic (e.g., as separate Parts) within a single lower layer stream.
- the source Agent is required to perform Part segmentation during encapsulation: it should separate different Parts with clear boundaries, and embed supplementary metadata (e.g., traffic information, QoS requirements) corresponding to each Part within the packet header or sub-headers. This supplementary metadata enables the underlying network to identify assistance information on specific Parts (e.g., their different traffic characteristics and respective QoS requirements).
    a) **In one implementation,** supplementary metadata of each Part can be encapsulated in the packet header (e.g., Agent protocol header, or lower layer header) as in Figure 4.

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AI Agent protocol for multi-modal Agent traffic

```
                    +-----------------------+
                    |Sub-header{            |
                    |   Part ID / SN,       |
                    |   Traffic information, |
                    |   QoS requirements    |
                    |}                      |
                    +-----------------------+
                    |                      /
                    |                    /
                    |                  /
                    |                /
                    |              /
                    +------------------+  +------------------+
                    |Sub-header|Payload|  |Sub-header|Payload|
                    +------------------+  +------------------+
                    |                  /  /                  /
                    |                 / /                  /
                    |                //                   /
+-------------------------------------------------------------+
|       Header      |               |                     |     |
| number of subPDUs |     subPDU    |       subPDU        | ... |
+-------------------------------------------------------------+
```

For example, each sub-header contains fields such as:

- sequence number of each Part and the corresponding traffic information (e.g. size of a Part, modality of a Part, and transmission mode of a Part), QoS requirements (e.g., latency, error rate, priority, privacy), etc.

The contents of different Parts, on the other hand, will be encapsulated in the payload following the corresponding sub-header.

In this case, a common header (e.g., in front of the subPDUs) can contain the field to indicate the total number of Parts of an AI Agent traffic.

b) **In another implementation,** supplementary metadata of each Part can be encapsulated in the sub-header of the sub-PDU.

Upon data arrival at the receiving endpoint,

- the entity of underlying network can perform reassembly of data packets associated with an AI Agent traffic prior to delivery to the receiving Agent client/server. Or,
- the entity of Agent protocol layer in the receiving Agent client/server can reassemble the data packets.

Which one is better can be left for further study.

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Summary

- The challenge wherein the underlying network lacks awareness of multi-modality of AI Agent traffic.

- Two different ways for the source Agent client or server to encapsulate multi-modal data:
  1. one is to encapsulate multi-modal data into multiple streams,
  2. another one is to encapsulate multi-modal data into a single stream.
  - In the former case, the source Agent notifies stream correlation information to the underlying network.
  - In the latter case, the source Agent separates multi-modal data into different Parts, and embed supplementary metadata (e.g., traffic information and QoS requirements) within the header and/ or sub-headers of Agent protocol layer or lower layers.

- For both methods, upon receiving the stream(s), underlying network can apply differential treatment to individual Parts, e.g., based on their specific QoS requirements.

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